Yashavantha Gowda Y D

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Profile

Embedded Software Developer with 4+ years of experience in firmware development, BSP integration, and driver programming for automotive and IoT platforms. Skilled in embedded C, ARM processors, communication protocols , RTOS (VxWorks), bootloaders, and debugging tools.

Technical Skills & Core Competencies

Programming Languages — C, Embedded C, shell scripting

OS & Kernel — Embedded Linux (BSP, device drivers, IIO framework), VxWorks 7 (BSP, CDF, DTS), U-Boot

SoCs & Hardware — Qualcomm SA8255P / SA7255P, TI AM62x LP Evaluation board, ARM Cortex-A53/M4

Bootloader — U-Boot bring-up, defconfig/DTS customization, OSPI boot, UART/USB DFU/TFTP boot, boot time optimization

Protocols — I2C, SPI, UART, QUPv3 Serial Engines, DDL1 (Diagnostic Data Link System)

RTOS - VxWorks7 — Board Support Package | Device tree source/DTSI files | Component Description language(CDF Files) | Device drivers

Version Control System — GIT,Gerrit

Professional Experience

KPIT Technologies — 05/2025 – Present

Software Engineer — Bangalore

Project : Skipgen4 Platform development

- Configured QUPv3 Serial Engines (I2C/SPI/UART) for Qualcomm SA8255P/ SA7255P in TrustZone, enabling secure peripheral access across Linux (PVM) and Android (GVM) virtual machines..
- Built and managed CDT binaries for multi-platform initialization, ensuring correct board configuration and peripheral accessibility.
- Developed and validated an I2C-based magnetometer driver (MMC5983MA) using the Linux IIO framework, adding triggered buffer support and software-based periodic sampling.
- Migrated and validated QUP test utilities across platforms, ensuring correct Serial Engine configuration for different hardware revisions.
- Ported and integrated MFi test utility into Yocto build system, verifying I2C communication and device identification

Tata Consultancy Services — 2021 – 05/2025

System Engineer

Project : Vehicle Diagnostic Device (OBD Device)

- Performed board bring-up for TI AM62x by building and configuring U-Boot, customizing defconfig and device tree for multiple boot modes.
- Implemented boot table management in OSPI boot mode, handling the storage and validation of bootable images such as uVxWorks and .dtb. Integrated CRC checks to ensure integrity during boot, all under Linux U-Boot environment.

- Focused on boot time optimization, including disabling unused features in U-Boot, reducing console output, and minimizing boot delay.
- Developed and automated boot scripts to seamlessly copy VxWorks OS images into RAM, ensuring automatic system boot upon power-up.
- Worked on GPIO device control (e.g., LED blinking, buzzer,trigger switch) under U-Boot for hardware validation before handing off to the VxWorks kernel.
- Implemented UART communication (DDL1 protocol) and developed low-level UART driver code for Bluetooth (BT122) and other modules, ensuring robust data transmission.
- Experienced in embedded C programming, low-level driver development, hardware abstraction layers, and firmware debugging for automotive and IoT devices.

Tools

VxWorks IDE 24.03 , Code Composer studio(CCS), Visual Studio, Gerrit, IBM rational Tools(Doors), Trace32.

Education

BE : Electrical and Electronics Engineering Mysore
JSS Science & Technology University Mysore.

Courses

Embedded development

- Completed Embedded Systems course at Emertxe, Bangalore, gaining expertise in C Programming, RTOS, and Embedded System design and development.
- Attended VxWorks driver development training given by Wind River team.